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ACADEMICS

Curriculum – Chemical Engineering

The Chemical Engineering curriculum at IIT Hyderabad is structured to give you a strong foundation in core sciences and engineering, followed by advanced departmental courses, laboratories, electives and project work. Where more than one revision exists for a programme, use the dropdown to switch between them.

Programme

B.Tech

B.Tech in Chemical Engineering

2025–Present ▾

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Curriculum for the 2025 batch onwards (127 credits). Restructures the earlier programme with Biological Engineering, an Artificial Intelligence course, and a Departmental Elective/Free Elective split moved into semesters V–VIII, alongside the standard basic sciences, basic engineering, core chemical engineering and liberal arts courses.

Credit Structure

Core	54 credits
Basic Engineering	16 credits
Basic Sciences	16 credits
Liberal Arts	10 credits
Departmental Elective	18 credits
Free Elective	13 credits
Total	127 credits

Progression by Semester

- Semesters 1–2:** Basic sciences, programming, applied chemistry, thermodynamic fundamentals and communication skills.
- Semesters 3–4:** Biological engineering, transport phenomena, thermodynamics, applied mathematics, reaction engineering, heat transfer and fluid mechanics.
- Semesters 5–6:** Mass transfer, mechanical operations, chemical technology, process control, unit-operations and reaction-engineering labs, plus the first departmental and free electives.
- Semesters 7–8:** Process design and economics, process simulation and control labs, process intensification, ethics, and remaining departmental/free electives.

Semester-wise Courses

Semester I			Total: 16 credits
Category	Code	Course	Credits
Basic Sciences	EP1108	Modern Physics	2
Basic Sciences	CY1010/CY1018	Environmental Chemistry	2
Basic Sciences	MA1110	Calculus-I	1
Basic Sciences	MA1220	Calculus-II	1
Basic Sciences	CH1120	Applied Chemistry	2
Basic Sciences	BT1010	Introduction to Life Sciences	1
Basic Engineering	ID1063	Introduction to Programming	3
Basic Engineering	CH1110	Introduction to Chemical Engineering	2
Liberal Arts	LA1760	Communication Skills	2

Semester II			Total: 15 credits
Category	Code	Course	Credits
Basic Sciences	CY1031	Chemistry Lab	2
Basic Sciences	MA1140	Elementary Linear Algebra	1
Basic Sciences	MA1150	Differential Equations	1
Basic Sciences	CH1131	Applied Chemistry Lab	1
Basic Engineering	ID1054	Digital Fabrication	2
Basic Engineering	CH1140	Thermodynamic Laws & Phase Transitions	3
Core	CH1130	Chemical Process Calculations	2
Liberal Arts	LAxxxx	LA/CA	3

Semester III			Total: 17 credits
Category	Code	Course	Credits
Core	CH2110	Biological Engineering	3
Core	CH2120	Numerical Methods	3
Core	CH2130	Transport Phenomena	3
Core	CH2140	Chemical Engineering Thermodynamics	3
Core	CH2150	Applied Mathematics in Chemical Engineering	3
Liberal Arts	LAxxxx	LA/CA	2

Semester IV			Total: 19 credits
Category	Code	Course	Credits
Basic Sciences	EP1031	Physics Lab	2
Basic Engineering	ID1050	Artificial Intelligence	1
Basic Engineering	EE1102	Basic Electrical Engineering	3
Basic Engineering	CH2160	Materials Science for Chemical Engineers	2
Core	CH2170	Chemical Reaction Engineering-I	3
Core	CH2180	Heat Transfer	3
Core	CH2190	Fluid Mechanics	3
Liberal Arts	LAxxxx	LA/CA	2

Semester V			Total: 19 credits
Category	Code	Course	Credits
Core	CH3110	Mass Transfer – I	3
Core	CH3120	Mechanical Operations	3
Core	CH3130	Chemical Technology	2
Core	CH3140	Chemical Reaction Engineering-II	3
Core	CH3121	HT & FM Lab	2
Departmental Elective	CHxxxx	Dept. Electives	6

Semester VI			Total: 17 credits
Category	Code	Course	Credits
Core	CH3150	Mass Transfer – II	2
Core	CH3160	Process Control	3
Core	CH3131	MUO and CRE Lab	2
Departmental Elective	CHxxxx	Dept. Elective	6
Free Elective	–	Free Electives-I	4

Semester VII			Total: 13 credits
Category	Code	Course	Credits
Core	CH4112	Process Design and Economics	3
Core	CH4111	Process Simulation Lab	2
Core	CH4121	MT and Control Lab	2
Departmental Elective	CHxxxx	Dept. Electives	3
Free Elective	–	Free Elective-II	3

Semester VIII			Total: 11 credits
Category	Code	Course	Credits
Core	CH4140	Process Intensification	1
Liberal Arts	ID4006	Ethics and Values	1
Departmental Elective	CHxxxx	Dept. Electives	3
Free Elective	–	Free Electives-III	6

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